

conditional statements in python

conditional statements in python is used to execute certain block of code if specific condition is true

if conditional statement

if statement is used to test the condition

if the condition is true the block of code inside the if runs

if the condition is false it skips that block 

```
In [1]: ▶ if True:  
        print('Data science')
```

Data science

```
In [2]: ▶ if True:  
        print('ds')
```

ds

```
In [3]: ▶ if False:  
        print('Data science')  
        print('bye for now')
```

bye for now

```
In [4]: ▶ if True:  
        print('Data science')  
        print('bye for now')
```

Data science
bye for now

```
In [5]: ▶ age=19  
        if age>18:  
            print('Eligible to vote')
```

Eligible to vote

if...else statement is used when the program allows to execute one block of code if condition is true and executes different block of code if condition is false

```
In [6]: ▶ if True:
        print('Data science')
        else:
        print('bye for now')
```

Data science

```
In [7]: ▶ age=10
        if age<=12:
            print('Travel for free')
        else:
            print('Pay for ticket')
```

Travel for free

short Hand if-else

short hand if-else statement allow us to write a single line if else statement

```
In [8]: ▶ marks=45
        res='pass' if marks>=40 else "Fail"
        print(f'Result:={res}')
```

Result:=pass

```
In [9]: ▶ x=4
        r=x%2
        if r==0:
            print('Even number')
```

Even number

```
In [10]: ▶ x=5
        r=x%2
        if r==0:
            print('Even number')
```

```
In [11]: ▶ x=5
        r=x%2
        if r==0:
            print('Even number')
        else:
            print('odd number')
```

odd number

```
In [12]: x=4
r=x%2
if r==0:
    print('Even number')
if r==1:
    print('odd number')
if r==2:
    print('even number')
```

Even number

```
In [13]: x = 1

if x == 1:
    print('one')
if x == 2:
    print('Two')
if x == 3:
    print('Three')
if x == 4:
    print('four')
```

one

```
In [14]: x=5
r=x%2
if r==0:
    print('Even number')
else:
    print('odd number')
```

odd number

Nested if else

Nested if else means an if else statement inside the if statement. We can use nested if statement to check conditions within conditions

```
In [15]: x=3
r=x%2
if r==0:
    print('Even number')
    if x>5:
        print('greater number')
else:
    print('odd number')
```

odd number

```
In [16]: x=4
r=x%2
if r==0:
    print('Even number')
    if x>5:
        print('Greater number')
else:
    print('Odd number')
```

Even number

```
In [17]: x=6
r=x%2
if r==0:
    print('Even number')
    if x>5:
        print('Greater number')
else:
    print('Odd number')
```

Even number

Greater number

```
In [18]: x=6
r=x%2
if r==0:
    print('Even number')
    if x>5:
        print('Greater number')
    else:
        print('lesser number')
else:
    print('Odd number')
```

Even number

Greater number

```
In [19]: x=4
r=x%2
if r==0:
    print('Even number')
    if x>5:
        print('Greater number')
    else:
        print('lesser number')
else:
    print('Odd number')
```

Even number

lesser number

```
In [26]: ▶ age=70
is_member=True
if age>=60:
    if is_member:
        print("30% senior discount!")
    else:
        print('20% senior discount')
else:
    print("Not eligible for a senior discount")
```

30% senior discount!

```
In [20]: ▶ x=4
if(x==1):
    print('one')
elif(x==2):
    print('two')
elif(x==3):
    print('three')
elif(x==4):
    print('four')
```

four

elif it wont check till the block if it find the output

```
In [21]: ▶ x=4
if(x==4):
    print('one')
elif(x==2):
    print('two')
elif(x==3):
    print('three')
elif(x==4):
    print('four')
```

one

```
In [22]: ▶ x=4
if(x==1):
    print('one')
else:
    print('two')
```

two

```
In [23]: x=4
if(x==4):
    print('one')
if(x==2):
    print('two')
if(x==3):
    print('three')
if(x==4):
    print('four')
```

one
four

```
In [24]: x=7
if(x==1):
    print('one')
elif(x==2):
    print('Two')
elif(x==3):
    print('Three')
elif(x==4):
    print('four')
```

```
In [25]: x=7
if(x==1):
    print('one')
elif(x==2):
    print('Two')
elif(x==3):
    print('Three')
elif(x==4):
    print('four')
else:
    print('number not found')
```

number not found

```
In [27]: score=90
if score>75:
    print("pass")
else:
    print('fail')
```

pass

```
In [31]: ▶ score=76
          destinction=90
          if score>=75:
              print("pass")
              if destinction==75:
                  print("gold medal")
              else:
                  print("no medal")
          else:
              print("fail")
```

pass
no medal

```
In [33]: ▶ age=18
          if((age>=8) and (age<=12)):
              print("you are allowed welcome")
          else:
              print("sorry ! you are not allowed")
```

sorry ! you are not allowed

```
In [35]: ▶ var='N'
          if(var=='Y' or var=="Y"):
              print('you said yes')
          elif(var=='N' or var=='n'):
              print("you said no")
          else:
              print("Invalid input")
```

you said no

```
In [42]: ▶ a=7
          b=9
          c=3
          if((a>b and a>c) and (a!=b and a!=c)):
              print(a,"is the largest")
          elif((b>a and b>c) and (b!=a and b!=c)):
              print(b,"is the largest")
          elif((c>a and c>b) and (c!=a and c!=b)):
              print(c,"i the largest")
          else:
              print("entered numbers are equal")
```

9 is the largest

```
In [43]: ▶ a=1
          b=1
          c=1
          if(a==1 and b==1 and c==1):
              print("working")
          else:
              print("stopped")
```

working

```
In [44]: ▶ a=20
          b=True
          if a>=18 and b:
              print("Eligible")
          else:
              print("not eligible")
```

Eligible

```
In [45]: ▶ a=20
          b=False
          if a>=18 and b:
              print("Eligible")
          else:
              print("not eligible")
```

not eligible

```
In [46]: ▶ a=23
          b="yes"
          if a>=18 and b=="yes":
              print("Granted")
          else:
              print("Denied")
```

Granted

```
In [47]: ▶ a=23
          b="no"
          if a>=18 and b=="yes":
              print("Granted")
          else:
              print("Denied")
```

Denied


```
In [49]: ▶ p="securePass123"
if len(p) >=8 and any(char.isdigit() for char in p):
    print("valid")
else:
    print("Invalid")
```

valid

```
In [9]: ▶ a=1
b=-2
flag=False
if((a>=0) ^ (b>=0)) and not flag:
    print("True")
elif((a<=0) ^ (b<=0)) and flag:
    print("True")
else:
    print("False")
```

True

```
In [21]: ▶ a = 1
b = -2
flag = True

if ((a >= 0) ^ (b >= 0)) and not flag:    # XOR check when flag is False
    print("True")
elif ((a < 0) ^ (b < 0) )and flag:        # both negative when flag is True
    print("True")
else:
    print("False")
```

True

```
In [15]: ▶ a = 1
b = -1
flag = False

if ((a >= 0) ^ (b >= 0)) and not flag:    # Case 1: exactly one non-negative,
    print("True")
elif (a < 0 and b < 0) and flag:          # Case 2: both negative, flag True
    print("True")
else:
    print("False")
```

True

```
In [24]: ▶ str="catinahat"
s1=str.count("cat")
s2=str.count("hat")
if(s1==s2):
    print("true")
else:
    print("false")
```

true

```
In [3]: ▶ str=" bazingaa"
s1=str.count("cat")
s2=str.count("gaa")
if(s1==s2):
    print("true")
else:
    print("false")
```

false

```
In [4]: ▶ a=10
if a>100:
    print("Big")
else:
    print("Number")
```

Number

```
In [11]: ▶ a=3
if a%3==0:
    print("Fizz")
elif a%5==0:
    print("Buzz")
elif (a%3==0 & a%5==0):
    print("FizzBuzz")
else:
    print(a)
```

Fizz

```
In [18]: ▶ a=15
if a%3==0:
    print("Fizz")
elif a%5==0:
    print("Buzz")
elif (a%3==0 and a%5==0):
    print("FizzBuzz")
else:
    print(a)
```

Fizz

```
In [25]: ▶ a=int(input("Enter a number"))
if a%3==0 and a%5==0:
    print("FizzBuzz")
elif a%3==0:
    print("Fizz")
elif a%5==0 :
    print("Buzz")
else:
    print(a)
```

Enter a number10
Buzz

```
In [4]: ▶ n=9
if n%2==1:
    print("you")
else:
    print("Friend")
```

you

```
In [6]: ▶ n=15
if n%2==0:
    print("true")
else:
    print("false")
```

false

```
In [15]: ▶ a=1
b=7
c=5
if (c>a and c>b):
    print("c is greatest")
elif(a>b and a>c):
    print("a is greatest")
else:
    print("b is greatest")
```

b is greatest

```
In [16]: ▶ a=1
b=7
c=5
print(max(a,b,c))
```

7

```
In [26]: ▶ year=2022
if(year%4==0 and year%100!=0) or (year%400==0):
    print("true")
else:
    print("false")
```

false

```
In [19]: ▶ year=2022
if year%4==0:
    print("true")
```

```
In [24]: ▶ year=2022
if year%100!=0:
    print("true")
```

true

```
In [25]: ▶ year=2022
if year%400==0:
    print("true")
```

```
In [4]: ▶ a=1
b=2
operator=int(input("enter an operator:"))
if operator==1:
    print(a+b)
elif operator==2:
    print(a-b)
elif operator==3:
    print(a*b)
else:
    print("Invalid input")
```

enter an operator:6
Invalid input

```
In [5]: ▶ n=13
m=4
num=int(input("Enter a number"))
if abs(num)==n and num%m==0:
    print("closest number")
else:
    print("not closest")
```

Enter a number12
not closest

```
In [ ]: ▶
```

